Alan Whelan

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Personal Statement

Natural leader, demonstrated through years of volunteer and work experience. Self-directed with the ability to learn quickly and with exceptional discipline. Passionate for physical oceanography, meteorology, instrumentation, and library sciences. Observant with a keen focus on details wihle maintaining the bigger picture. Maintains an optimistic viewpoint in all aspects of work and life.

Education

University of Maryland, College Park — College Park, MD
B.S. in Atmospheric and Oceanic Science — *2024 – 2026*

Frederick Community College — Frederick, MD
A.S. in STEM, concentrations in Biology, Physics — *2020 – 2023*

Relevant Coursework

- Atmospheric Thermodynamics
- Calculus I, II, & III
- Chesapeake Bay Oceanography
- Climate Data Science
- Climate Dynamics and Earth System Science
- Dynamics of the Atmosphere and Ocean
- General Physics I, II, & III
- Machine Learning in Earth Science
- Organic Chemistry I & II
- Partial Differential Equations
- Physical Oceanography
- Remote Sensing of the Atmosphere and Ocean
- Scientific Data Visualization with Python
- Synoptic and Mesoscale Meteorology
- Weather & Climate

Awards and Scholarships

Jeffrey and Lily Chen Scholarship *2024 – 2026*
University of Maryland and NSF Chesapeake Scholars S-STEM Scholarship *2024 – 2026*
Dr. Patricia Stanley Excellence in Leadership Award *2022*
Maryland Association of Community Colleges Student Advocacy Day Speaker *2021*
Frederick Community College and NSF S-STEM Scholarship *2020 – 2022*

Posters and Presentations

Spatial and Temporal Trends in Southern Ocean Eddy Kinetic Energy *2025-2026*

- Senior thesis research project, advised by Dr. Madeleine Youngs at the University of Maryland.
- Presented at the University of Maryland's Burgers Fluid Dynamics Symposium and at their Department of Atmospheric and Oceanic Science's Senior Research Symposium.

Investigating the variability of the Good Hope Jet *2024*

- Advised by Lindsay Grose and Dr. Kathy Donohue at the University of Rhode Island's Graduate School of Oceanography.
- Presented at the end-of-summer research symposium with fellow interns.

On Learning the Aldol Reaction by Synthesizing Indigo *2022*

- Completed for Honors Organic chemistry II, presented at the Frederick Community College Honors Forum.

On Possible Solutions for Collapsing Coral Reef Ecosystems *2021*

- Presented at the Northeast Regional Honors Conference

Leadership Experience

American Meteorological Society Student Chapter at UMD — College Park, MD
Treasurer — *May 2024 – 2026*

- Attended and contributed to biweekly science education meetings.
- Balanced organization's budget and filed expenses.
- Represented University of Maryland at yearly American Meteorological Society meetings.
- Participated in educational outreach, including at the Maryland Day research showcase.

Frederick Community College Student Government Association — Frederick, MD
Representative to the College Senate — *Jun 2022 – Jun 2023*

- Attended and voted in internal college governance meetings on behalf of the Student Government Association.
- Participated in college policy creation and review through recommendations.
- Collaborated with college leadership to ensure the Student Government's thousands of constituents were represented within all college governance processes.

Maryland Higher Education Commission — Baltimore, MD
Vice Chair of the Student Advisory Council — *Oct 2021 – Jun 2022*

- Assisted in supervision and operation of Student Advisory Council meetings of approximately twenty students.
- Worked with student leaders from other colleges to choose meeting topics, invite guest speakers, and create monthly meeting agendas.
- Contributed to the Council's Committee to Address the Silencing of Student Voices.

Frederick Community College Student Government Association — Frederick, MD
President — *Jun 2021 – Jun 2022*

- Led and contributed to weekly public and internal meetings of 20-30 students and college leaders.
- Created meeting agendas in coordination with fellow Student Government leadership.
- Listened and responded to student feedback through advocacy within the internal college structure.
- Assisted in the launch of the new Navigate registration and advising platform.
- Supported the search committee tasked with hiring a new President of the college through touring and evaluating candidates.

Work Experience

University of Maryland STEM Library — College Park, MD
Student Assistant — *Jan 2024 – May 2026*

- Searched for books placed on hold and processed materials proficiently in Ex Libris Alma.
- Assisted patrons with locating resources on STEM subjects using the Library of Congress classification.
- Verified materials were in order and reshelved as necessary, tracking progress on Excel spreadsheet and flagging incorrect call number labels.
- Scanned research articles and ensured OCR readiness for interlibrary loan services.

Makerspace Assistant — *Sep 2024 – Sep 2025*

- Operated and maintained 3D printers, vinyl cutters, and laser printers.
- Prepared, tracked, and completed 3D print project requests.
- Designed educational and engaging programming to demonstrate productive use of Makerspace equipment (Virtual Reality Visualization Demo).

Regal Cinemas — Frederick, MD
Floor Staff — *Jan 2020 – Nov 2020, Sep 2023 – Jan 2024*

- Served customers at the concession counter in a convenient, friendly, and timely manner.
- Cleaned facilities and auditoriums, including frequent sanitization.
- Greeted guests at the door, checked tickets, and directed guests to auditoriums.

Frederick Community College Library — Frederick, MD
Circulation Assistant, STEM Tutoring Assistant — *Jun 2023 – Jan 2024*

- Assisted patrons with printing and copying, checking out books, and checking in for tutoring appointments.
- Took inventory of the thousands of books in the library's collection.
- Completed data entry projects.
- Assisted in archival of hundreds of old documents pertaining to the history of the college.

Technical Skills

- Soldering of surface-mount and through-hole electronics.
- Disassembly, repair, and maintenance of small instrumentation and electronics.
- Use of Microsoft Office applications to generate PowerPoint presentations, Word documents, and Excel spreadsheets.
- Transformation and aggregation of large datasets, including HDF5, NetCDF4, npy, and CSV formats.
- Programming to create plots and animations to visualize datasets in MATLAB, Python/Matplotlib, and ParaView.
- Proficient use of Linux terminal and Windows command line.

Additional Projects

Personal website *2024-2026*

- Created HTML/CSS/Js website from scratch to display results of research projects.
- Iterated on over years and transferred from University department site to Github page.

Virtual Reality Data Visualization Demo *2025*

- This project was completed for the University of Maryland STEM Library's Makerspace, as an interactive demonstration of the available virtual reality equipment.
- This project was designed after a demo of NASA's Mixed Reality Exploration Toolkit, but instead used Paraview and its VR plugin to display ocean model data in Virtual Reality.
- The demo and poster were presented in the Makerspace as an event, reaching approximately 20 attendees.

Data Analysis for UMD Balloon Payload Program *2024*

- Combed through past radiosonde data collected by UMD student group to create Skew-T plots in Python.
- Worked around unlabeled data, instrument error, and lack of reliable humidity measure to produce mock-up Skew-Ts and flight path animations.

Live plots of Maryland Mesonet Station Data *2024*

- Connected Python to SQL database to pull in five-minutely weather station measurements and plot them on their locations on a map of Maryland.
- This project was an exercise in live data collection and how processes need to be run in order to periodically update a live web view.